

# BIZEV-05: KPIs and Metrics

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**Series:** BIZEV — Business Events & Funnel Analysis | **Notebook:** 5 of 7 | **Created:** March 2026 | **Last Updated:** 09/18/2026

## Overview

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Business events contain the raw signals needed to compute meaningful Key Performance Indicators (KPIs) — transaction volume, average order value, error rates by business process, and more. This notebook shows how to define and compute business KPIs from event data using DQL, extract metrics from business events via OpenPipeline for long-term retention, build composite business health scores, and perform trend analysis to detect shifts in business performance.

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# Prerequisites

| Requirement           | Details                                                                               |
|-----------------------|---------------------------------------------------------------------------------------|
| Dynatrace Environment | SaaS with Grail enabled                                                               |
| Permissions           | <code>storage:bizevents:read</code> , <code>storage:metrics:read</code>               |
| Data                  | Business events with transaction types and value fields                               |
| Knowledge             | BIZEV-01 through BIZEV-03, DQL <code>summarize</code> and <code>makeTimeseries</code> |

## 1. Defining Business KPIs

A KPI is a quantifiable measure of business performance. The best KPIs are:

| Property   | Description                         |
|------------|-------------------------------------|
| Specific   | Tied to a concrete business outcome |
| Measurable | Calculable from available data      |
| Actionable | Teams can influence the metric      |
| Time-bound | Measured over a defined period      |

## Common Business Event KPIs

| KPI                       | Formula                                    | Business Question              |
|---------------------------|--------------------------------------------|--------------------------------|
| Transaction Volume        | <code>count ( )</code> of order events     | How many orders per hour/day?  |
| Average Order Value (AOV) | <code>avg (amount )</code> of order events | How much per order on average? |
| Conversion Rate           | Orders / Views * 100                       | What % of visitors buy?        |
| Error Rate                | Failed / Total * 100                       | What % of transactions fail?   |

| KPI              | Formula                           | Business Question                    |
|------------------|-----------------------------------|--------------------------------------|
| Revenue per Hour | <code>sum(amount)</code> per hour | What is the hourly revenue run rate? |
| Time to Complete | Avg time from start to end event  | How long does a transaction take?    |

## 2. Transaction Volume KPIs

Transaction volume is the most fundamental business KPI — it tells you how active your business is.

```
// Transaction volume by event type over the last 24 hours
fetch bizevents, from:-24h
| summarize volume = count(), by:{event.type}
| sort volume desc
| limit 15
```

```
// Hourly transaction volume trend – spot peak and off-peak hours
fetch bizevents, from:-24h
| makeTimeseries tx_volume = count(), interval:1h
```

```
// Daily transaction volume for the past 7 days by event type
fetch bizevents, from:-7d
| fieldsAdd day = bin(timestamp, 1d)
| summarize daily_volume = count(), by:{day, event.type}
| sort day asc, daily_volume desc
```

## 3. Revenue KPIs

Revenue KPIs require an `amount` or `total` field in your business events. These are the metrics executives care about most.

**Note:** Replace `amount` with the actual field name in your business events that carries the monetary value.

```
// Revenue KPI summary for the last 24 hours
fetch bizevents, from:-24h
| filter event.type == "com.myapp.order.completed"
| filter isNotNull(amount)
| summarize {total_revenue = sum(toDouble(amount)),
              avg_order_value = avg(toDouble(amount)),
              median_order_value = median(toDouble(amount)),
              max_order_value = max(toDouble(amount)),
              order_count = count()}
```

```
// Revenue timeseries – hourly revenue over the last 24 hours
fetch bizevents, from:-24h
| filter event.type == "com.myapp.order.completed"
| filter isNotNull(amount)
| makeTimeseries hourly_revenue = sum(toDouble(amount)), interval:1h
```

```
// Average order value trend – daily for the past week
fetch bizevents, from:-7d
| filter event.type == "com.myapp.order.completed"
| filter isNotNull(amount)
| fieldsAdd day = bin(timestamp, 1d)
| summarize {aov = avg(toDouble(amount)),
              daily_revenue = sum(toDouble(amount)),
              orders = count()}, by:{day}
| sort day asc
```

## 4. Error Rate KPIs

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Business process error rates measure the percentage of transactions that fail. This differs from HTTP error rates — a payment can fail (business error) even when the HTTP call succeeds (200 OK).

```
// Business error rate by event type
// Assumes failed events have a status or result field
fetch bizevents, from:-24h
| filter in(event.type, {"com.myapp.payment.processed",
                        "com.myapp.payment.failed"})
| summarize {total = count(),
              failures = countIf(event.type == "com.myapp.payment.failed")}
| fieldsAdd error_rate_pct = round(toDouble(failures) / toDouble(total) * 100,
                                   decimals: 2)
```

```
// Error rate trend over time – detect spikes
fetch bizevents, from:-24h
| filter in(event.type, {"com.myapp.payment.processed",
"com.myapp.payment.failed"})
| makeTimeseries total = count(),
                  failures = countIf(event.type == "com.myapp.payment.failed"),
                  interval:1h
```

```
// Error rate by provider – which channels have the most failures?
fetch bizevents, from:-24h
| filter in(event.type, {"com.myapp.payment.processed",
"com.myapp.payment.failed"})
| summarize {total = count(),
              failures = countIf(event.type == "com.myapp.payment.failed")}, by:
{event.provider}
| filter total > 10
| fieldsAdd error_rate_pct = round(toDouble(failures) / toDouble(total) * 100,
decimals: 1)
| sort error_rate_pct desc
```

## 5. Metric Extraction via OpenPipeline

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While DQL queries compute KPIs on demand, **OpenPipeline metric extraction** creates persistent Dynatrace metrics from business events. This enables:

- Long-term retention (metrics have longer default retention than events)
- Alerting via Dynatrace Intelligence or custom metric events
- Dashboard widgets with instant load times
- SLO definitions based on business metrics

# OpenPipeline Configuration

```
# Example: Extract order count and revenue as metrics
pipelines:
  - name: "Business KPI Metrics"
    processing:
      - type: metric_extraction
        condition: "event.type == 'com.myapp.order.completed'"
        metrics:
          - key: "business.orders.count"
            type: count
          - key: "business.orders.revenue"
            type: gauge
            value: "amount"
            dimensions:
              - event.provider
              - product_category
```

## Benefits of Metric Extraction

| Approach          | Latency    | Retention                          | Alerting                          | Cost             |
|-------------------|------------|------------------------------------|-----------------------------------|------------------|
| DQL on bizevents  | Seconds    | Event retention (default 35 days)  | Manual threshold                  | Scans event data |
| Extracted metrics | Sub-second | Metric retention (default 5 years) | Native Dynatrace Intelligence/SLO | Pre-aggregated   |

**Classic-pipeline variant — the bridge to classic dashboards.** Alongside the OpenPipeline path shown here, Dynatrace documents business-event metric extraction via the classic pipeline (Settings > Business Observability > Metric extraction). Rules there emit metrics keyed with the `bizevents.` prefix (up to 50 dimensions) that are readable in Data Explorer, classic dashboards, and classic metric-event alerting — useful when report consumers still live in the classic (Gen2) experience. It is closing, though — per [Business event metric extraction via classic pipeline \(DT docs\)](#): *"For new accounts created from September 2026, classic pipeline is not available. Use OpenPipeline to process your data."* Build new extraction on the OpenPipeline path; either way the output is an ordinary metric. See **BIZEV-07: Gen2 vs Gen3 — Business Events Without (or Before) the Full Move** for the full hybrid-adoption pattern.

```
// If you have extracted business metrics, query them with timeseries
// Replace with your actual metric key
timeseries order_volume = count(business.orders.count), from:-24h
```

## 6. Business Health Score

A composite health score combines multiple KPIs into a single indicator. This simplifies executive reporting and enables at-a-glance health checks.

### Example Scoring Model

| KPI                               | Weight | Green             | Yellow | Red   |
|-----------------------------------|--------|-------------------|--------|-------|
| Transaction volume (vs baseline)  | 30%    | > 90% of baseline | 70-90% | < 70% |
| Error rate                        | 30%    | < 1%              | 1-5%   | > 5%  |
| Average order value (vs baseline) | 20%    | > 95% of baseline | 80-95% | < 80% |
| Conversion rate (vs baseline)     | 20%    | > 90% of baseline | 70-90% | < 70% |

```
// Compute a simple business health dashboard – key metrics at a glance
fetch bizevents, from:-1h
| summarize {total_events = count(),
              distinct_types = countDistinct(event.type),
              distinct_providers = countDistinct(event.provider)}
| fieldsAdd time_window = "Last 1 hour",
            status = if(total_events > 0, then: "ACTIVE", else: "NO DATA")
```

```
// Multi-KPI summary – compute all key metrics in one query
fetch bizevents, from:-24h
| summarize {total_transactions = count(),
              order_count = countIf(event.type == "com.myapp.order.completed"),
              payment_failures = countIf(event.type ==
"com.myapp.payment.failed"),
              total_payments = countIf(in(event.type,
{"com.myapp.payment.processed", "com.myapp.payment.failed"}))}
| fieldsAdd error_rate = if(total_payments > 0,
                           then: round(toDouble(payment_failures) /
toDouble(total_payments) * 100, decimals: 2),
                           else: 0.0)
| fieldsAdd health = if(error_rate < 1.0, then: "HEALTHY",
                       else: if(error_rate < 5.0, then: "DEGRADED", else:
"CRITICAL"))
```

## 7. Trend Analysis

Trends reveal whether KPIs are improving or degrading over time. Use `makeTimeseries` for visual trends and `summarize` with `bin()` for tabular comparisons.

```
// Weekly trend – transaction volume per day for the past 4 weeks
fetch bizevents, from:-28d
| fieldsAdd day = bin(timestamp, 1d)
| summarize daily_volume = count(), by:{day}
| sort day asc
```

```
// Week-over-week comparison using day-of-week grouping
fetch bizevents, from:-14d
| fieldsAdd week = getWeekOfYear(timestamp),
              dow = getDayOfWeek(timestamp)
| summarize volume = count(), by:{week, dow}
| sort week asc, dow asc
```

```
// Timeseries trend of business events by type – 7 days, 6h intervals
fetch bizevents, from:-7d
| makeTimeseries event_count = count(), by:{event.type}, interval:6h
```

## 8. Summary and Next Steps

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In this notebook, you learned:

- **Defining business KPIs** from event data — volume, revenue, error rates, conversion
- **Transaction volume analysis** — Counting, trending, and comparing event volumes
- **Revenue metrics** — AOV, hourly revenue, daily revenue trends
- **Error rate calculation** — Business-level failure rates distinct from HTTP errors
- **Metric extraction** — Using OpenPipeline to create persistent metrics from events
- **Health scoring** — Combining multiple KPIs into composite health indicators
- **Trend analysis** — Week-over-week and day-over-day comparisons

### Next Steps

- **BIZEV-06: Executive Reporting** — Package these KPIs into executive-ready reports
- Consider setting up OpenPipeline metric extraction for your most important KPIs

### References

- [Dynatrace Business Analytics](#)
- [OpenPipeline Metric Extraction](#)
- [DQL structuring commands \(DT docs\)](#)

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*This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.*